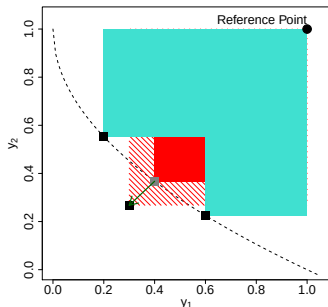
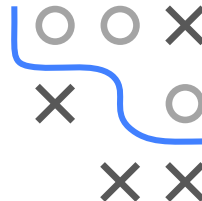


Optimization in Machine Learning

Bayesian Optimization

Multicriteria Bayesian Optimization

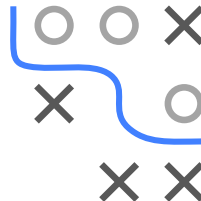
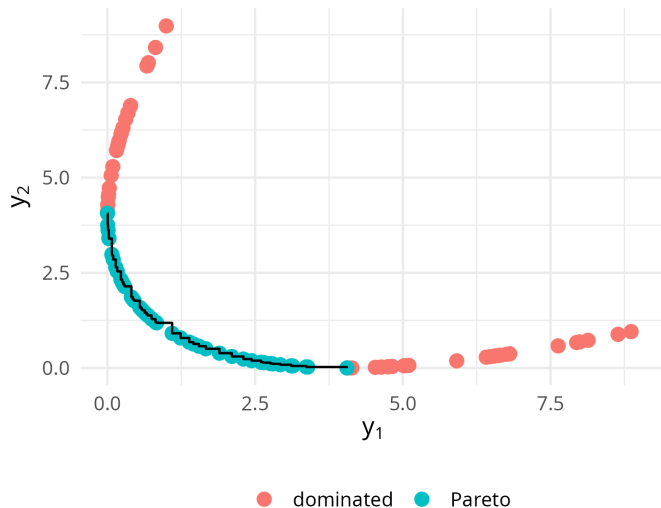


Learning goals

- Multicriteria Optimization
- Taxonomy
- ParEGO, SMS-EGO, EHI

MULTICRITERIA BAYESIAN OPTIMIZATION

Example Pareto front:



MULTICRITERIA BAYESIAN OPTIMIZATION

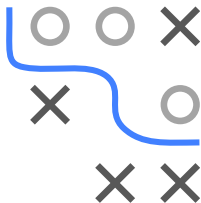
The most popular quality indicator is the hypervolume indicator (also called dominated hypervolume or $\mathcal{S}$ -metric).

The hypervolume, HV, of an approximation of the Pareto front $\hat{\mathcal{F}} = f(\hat{\mathcal{P}})$ can be defined as the combined volume of the dominated hypercubes domHC_r of all solution points $\mathbf{x} \in \hat{\mathcal{P}}$ regarding a reference point $\mathbf{r}$, i.e.,

$$\text{HV}_r(\hat{\mathcal{P}}) := \mu \left(\bigcup_{\mathbf{x} \in \hat{\mathcal{P}}} \text{domHC}_r(\mathbf{x}) \right)$$

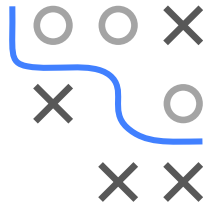
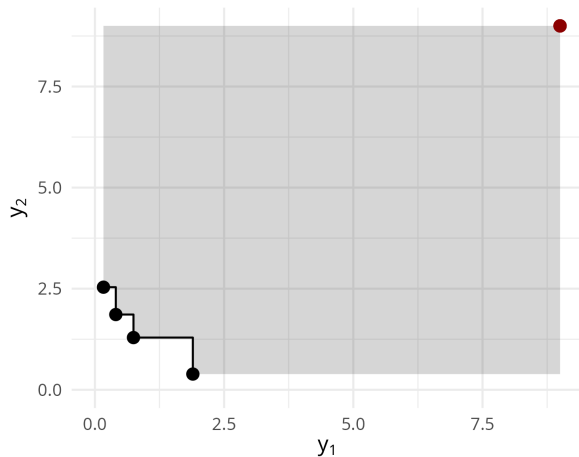
where μ is the Lebesgue measure and the dominated hypercube is given as:

$$\text{domHC}_r(\mathbf{x}) := \{ \mathbf{u} \in \mathbb{R}^m \mid f_i(\mathbf{x}) \leq \mathbf{u}_i \leq \mathbf{r}_i \forall i \in \{1, \dots, m\} \}$$



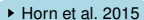
MULTICRITERIA BAYESIAN OPTIMIZATION

Hypervolume example:

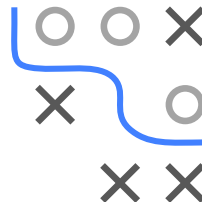
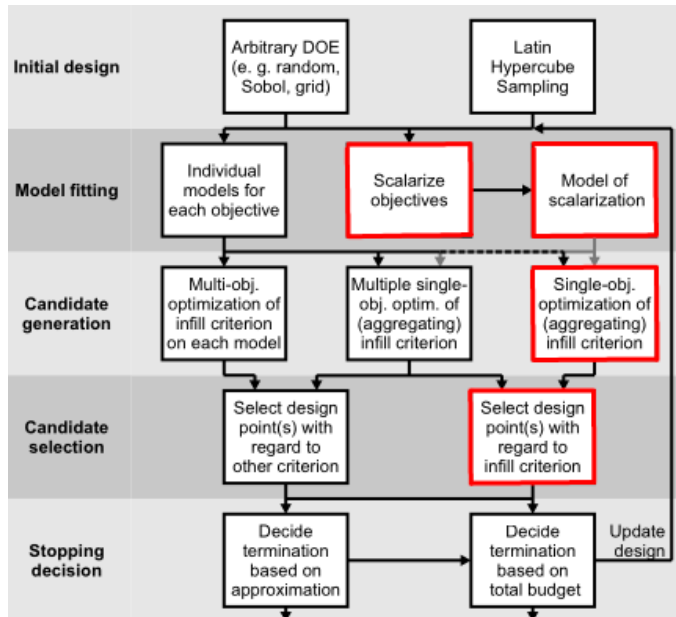


Reference point $\mathbf{r}$ in red, estimated Pareto front $\hat{\mathcal{F}}$ in black,
corresponding $HV_{\mathbf{r}}(\hat{\mathcal{P}})$ is given by the grey area

A 3x3 grid with a blue path starting at the top-left cell (0,0) and ending at the bottom-right cell (2,2). The path consists of the following cells: (0,0), (0,1), (1,1), (1,2), and (2,2). The cells (0,2), (1,0), and (2,0) are empty. The cells (1,0) and (2,0) contain a black 'X'. The cells (0,1) and (1,1) contain a grey circle. The cell (2,1) contains a grey circle.



PAREGO



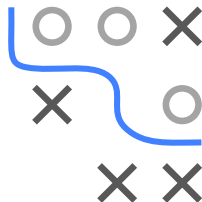
PAREGO

- 1 Scalarize standardized objectives using the augmented Tchebycheff norm

$$\max_{i \in \{1, \dots, m\}} w_i f_i(\mathbf{x}) + \rho \sum_{i=1}^m w_i f_i(\mathbf{x})$$

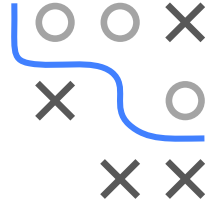
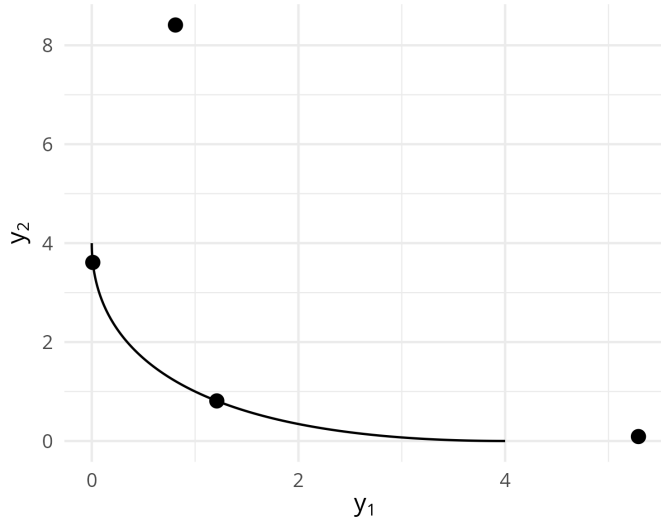
with weight vector $\mathbf{w}$ drawn uniformly from the set of evenly distributed weight vectors $\mathcal{W}$

- 2 Fit SM on the scalarized objective function
- 3 Proceed to use any standard single-objective acquisition function (EI, PI, LCB, ...)



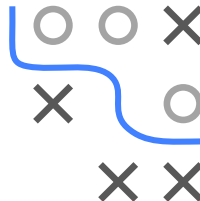
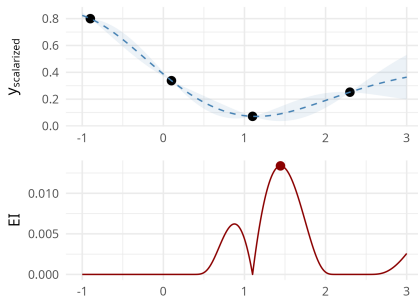
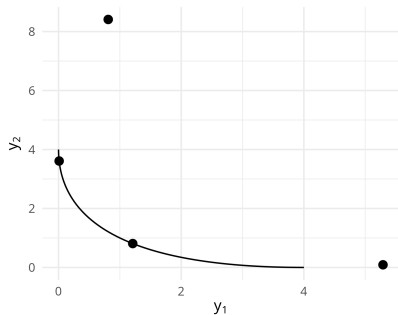
PAREGO

ParEGO Example, initial design and true Pareto front in black ...



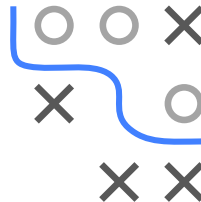
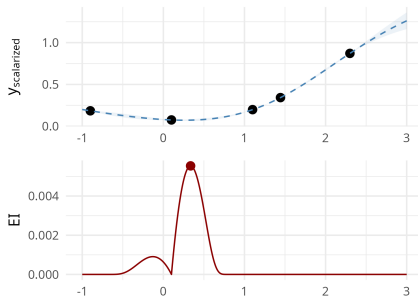
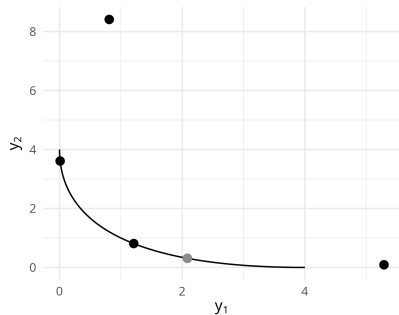
PAREGO

... standardize objectives, obtain scalarized objective via augmented Tchebycheff norm, fit SM and optimize EI ...



PAREGO

... note that the specific scalarization is different at each iteration!



The grey point visualizes the candidate we choose to evaluate in the previous iteration

```
graph TD
    subgraph Initial_design [Initial design]
        A[Arbitrary DOE  
(e. g. random,  
Sobol, grid)]
        B[Latin Hypercube Sampling]
    end

    subgraph Model_fitting [Model fitting]
        C[Individual models for each objective]
        D[Scalarize objectives]
        E[Model of scalarization]
    end

    subgraph Candidate_generation [Candidate generation]
        F[Multi-obj. optimization of infill criterion on each model]
        G[Multiple single-obj. optim. of (aggregating) infill criterion]
        H[Single-obj. optimization of (aggregating) infill criterion]
    end

    subgraph Candidate_selection [Candidate selection]
        I[Select design point(s) with regard to other criterion]
        J[Select design point(s) with regard to infill criterion]
    end

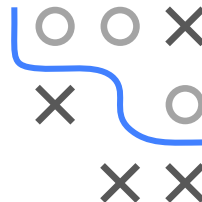
    subgraph Stopping_decision [Stopping decision]
        K[Decide termination based on approximation]
        L[Decide termination based on total budget]
    end

    A --> C
    A --> D
    B --> E
    C --> F
    D --> G
    D --> H
    E --> H
    F --> I
    G --> I
    G --> J
    H --> J
    I --> K
    J --> L
    K --> L
    L --> Update[Update design]
    Update --> B
```

The flowchart illustrates the multi-objective optimization process, organized into five main stages:

- Initial design:** Starts with two methods: "Arbitrary DOE (e. g. random, Sobol, grid)" and "Latin Hypercube Sampling".
- Model fitting:** The process branches into three parallel paths:
 - "Individual models for each objective" (highlighted with a red border)
 - "Scalarize objectives"
 - "Model of scalarization"
- Candidate generation:** Each path from model fitting leads to a corresponding optimization step:
 - "Multi-obj. optimization of infill criterion on each model" (from individual models)
 - "Multiple single-obj. optim. of (aggregating) infill criterion" (from scalarized objectives)
 - "Single-obj. optimization of (aggregating) infill criterion" (from model of scalarization, also receiving input from scalarized objectives via a dashed line)
- Candidate selection:** The optimization results lead to selection steps:
 - "Select design point(s) with regard to other criterion" (receiving input from multi-objective and multiple single-objective optimizations)
 - "Select design point(s) with regard to infill criterion" (highlighted with a red border, receiving input from multiple single-objective and single-objective optimizations)
- Stopping decision:** The final selection steps lead to termination decisions:
 - "Decide termination based on approximation" (from other criterion selection)
 - "Decide termination based on total budget" (highlighted with a red border, from infill criterion selection)

The process concludes with an "Update design" step, which feeds back into the "Latin Hypercube Sampling" method in the Initial design stage.

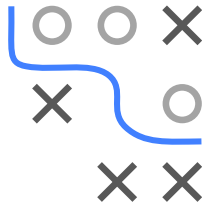
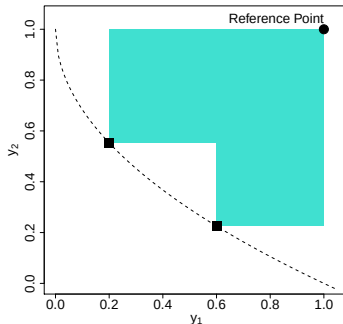


SMS-EGO

Individual models for each objective f_i

Single-objective optimization of aggregating acquisition function:
Calculate contribution of the confidence bound of candidate to the
current front approximation

- Calculate LCB for each objective
- Measure contribution with regard to the hypervolume improvement
- For ε -dominated ($\prec_\varepsilon$) solutions, a penalty is added

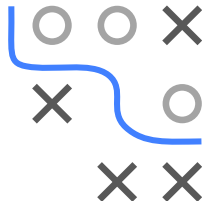
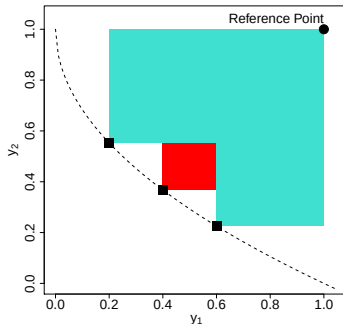


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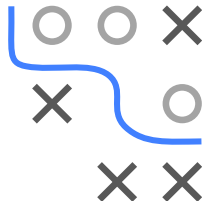
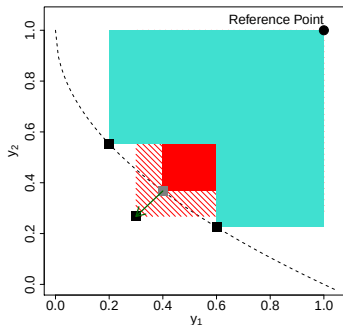


SMS-EGO

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OUTLOOK

- Many more options exist:
 - Expected Hypervolume Improvement
 - Multi-EGO
 - Entropy based: PESMO, MESMO
 - ...

